

KENNETH HERRERA

Senior AI Software Engineer · Agentic Systems & Cloud Infrastructure

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🔗 github.com/thequantpartners 📦 npm: [@quantpartners/pca](https://www.npmjs.com/package/@quantpartners/pca)

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience specializing in the **architecture, engineering, and production deployment of autonomous AI agent systems and resilient cloud infrastructure**. Creator of production-grade agentic gateways with **native Model Context Protocol (MCP - 27 tools)**, multi-LLM orchestration (Claude 3.5 Sonnet, GPT-4o, Gemini), deterministic Tool Calling, and multimodal RAG pipelines. Core expertise in **TypeScript/Node.js** and **Python**, multi-stage Docker containerization on Linux VPS, agent cybersecurity (Bot-to-Bot Loop Defense, Circuit Breakers), and high-throughput real-time messaging pipelines.

TECHNICAL SKILLS

Generative AI & Autonomous Agents: Model Context Protocol (Native MCP — 27 production tools over SSE and Stdio); Deterministic Tool Calling; Structured Outputs (Zod, Pydantic, JSON Schema); Advanced Prompt Engineering, anti-hallucination guardrails, and jailbreak defense; Multi-LLM routing (Claude 3.5 Sonnet, GPT-4o, Gemini 1.5 Flash); Multimodal RAG (150-300 page documents); Vector Stores (pgvector, PostgreSQL); Agentic Frameworks (MCP, LangChain).

Backend & Software Architecture: Modern TypeScript/Node.js (ESM, Express), Python (FastAPI, Pydantic v2); RESTful APIs, WebSockets/XMPP (Baileys), Asynchronous Webhooks, Server-Sent Events (SSE); Finite State Machines (FSM); Human-in-the-Loop (HITL) workflows; Transactional Idempotency and distributed queue processing.

Cloud, DevOps & Cybersecurity: Docker (Multi-stage builds, Alpine <180MB, layer cache optimization), Docker Compose; Linux VPS Hardening (Hetzner, Ubuntu Server); Self-hosted Coolify PaaS (Git-driven CI/CD); Cloudflare R2 (S3 API automated backup pipelines); Google Cloud Run, Railway, AWS SES. Bot-to-Bot Loop Defense, Circuit Breaker Pattern, AES-256 encryption, and zero-trust secret management.

Databases & Integrations: PostgreSQL (High-concurrency connection pools, ACID transactions, advanced indexing), Playwright (headless browser automation & scraping), Meta WhatsApp Cloud API, Resend Cloud API, Zoho Mail SMTP (STARTTLS / port 587), Outscraper / Apify APIs.

PROFESSIONAL EXPERIENCE

The Quant Partners — Lead AI Software Engineer & Co-Founder

2023 – Present | Remote (Lima, Peru)

- **Native MCP Server in Production:** Architected and deployed a centralized agentic gateway with 27 tools under the Model Context Protocol (MCP), enabling external LLM clients (Cursor, Claude Desktop, Antigravity) to execute autonomous B2B scraping, data analysis, and messaging in real time.
- **Autonomous 4-Agent Parallel Fleet:** Engineered a multi-agent ecosystem (Active Outbound Scraping, 24/7 Inbound Setter, Anti-Ghosting Reactivator, and CRM Sync) delivering sub-5s response latency with organic human presence emulation (dynamic typing simulation and cognitive reading pauses to bypass algorithmic telemetry).
- **Agent Cybersecurity & Fault Tolerance:** Implemented the *Bot-to-Bot Loop Defense* protocol utilizing Finite State Machines (FSM) and automated *Circuit Breakers*, preventing infinite ping-pong messaging loops with third-party chatbots and slashing wasted inference token spend by 40%.
- **Cloud Infrastructure & Continuous CI/CD:** Automated deployment pipelines via multi-stage Docker and Coolify on dedicated Linux VPS, ensuring transactional persistence in PostgreSQL and automated daily backups to Cloudflare R2 with 99.9% uptime.
- **Multimodal RAG Pipeline for Complex Documents:** Built a semantic document extraction engine with Gemini 1.5 Flash to parse complex legal and government bidding dossiers (150 to 300 pages), achieving zero false positives in technical compliance and penalty extraction.

Cloud Software Consultant & Backend Automation — Backend Developer

2022 – 2024 | Remote

- **Serverless Microservices & Web Automation:** Developed end-to-end automation pipelines for large-scale tax and regulatory data extraction using Node.js, Playwright, and Google Cloud Run with scalable serverless architecture.
- **RESTful APIs & Data Connectors:** Built transactional APIs with Node.js/Express integrating web platforms with payment gateways and messaging infrastructure, enforcing strict validation with Zod and PostgreSQL persistence.
- **Data Security & Encryption:** Implemented secure token-based authentication (JWT / API Keys), AES-256 sensitive data encryption, and centralized secret management in production environments.

Open Source Creator — @quantpartners/pca on npm

2025 – Present

- Authored and maintain a developer-facing CLI tool on npm designed to eradicate *context drift* in AI-assisted coding environments, featuring a local-first architecture and zero telemetry.

EDUCATION & LANGUAGES

B.S. in Systems and Software Engineering

Specialization: Agentic Architectures, Production GenAI, Distributed Systems & DevOps

Languages: English (Professional / Technical Fluent B2) | Spanish (Native)